

What Does a Routing Oracle Measure Under Stochastic Decoding? Coupling, Scorer Choice, and Single-Commit Ceilings

Teng-Ruei Chen

Abstract—Large language model routing benchmarks often compare a policy that commits to one model with a hindsight oracle that credits a query whenever any model’s recorded output is correct. Under stochastic decoding these are different decision classes, and the second object is not determined by per-model success probabilities unless a cross-model coupling is specified. For a frozen query–model–decoder–scorer protocol, let p_{im} be model m ’s success probability on query i . We distinguish the clairvoyant one-model ceiling $R_i = \max_m p_{im}$, a named product-coupling union $U_i^\perp = 1 - \prod_m (1 - p_{im})$, and the premium $\Delta_i^\perp = U_i^\perp - R_i$. With only these marginals fixed, the attainable union probabilities form exactly the interval $[R_i, \min\{1, \sum_m p_{im}\}]$; hence zero premium is attainable over couplings compatible with those marginals. If the protocol instead specifies independent generation across models, the product formula is its union probability, not an unidentified quantity.

We apply this measurement audit retrospectively to hash-bound correctness tensors from 11 open models, 30 archived responses per query–model cell, and GSM8K, MATH-500, and GPQA-Diamond. On the designated display channels, the full-pool empirical product premiums are 0.371, 3.542, and 5.100 percentage points, respectively. Their marginally identified premium intervals—whose zero lower endpoints follow algebraically rather than empirically—are $[0, 0.473]$, $[0, 4.787]$, and $[0, 7.744]$ points. Across every retained scorer, the corresponding ranges are 0.359–0.398, 3.542–3.754, and 5.100–6.632 points. Eight frozen finite- k paths and exact enumeration of all $2^{11} - 1$ nonempty model subpools expose estimator and pool sensitivity without turning either spread into a confidence interval. The designated GSM8K and GPQA display scorers were developed after limited output inspection and are labeled accordingly. An anonymous evidence bundle binds the frozen tensors, numerical contract, canonical Run-A outputs, and their hashes for independent number checks.

The main audit measures fixed archived tensors under named scorers and couplings, not a population effect or an equal-cost routing gain. A separate retrospective held-out policy illustration instantiates the decomposition; it does not establish new-data generalization. The practical contribution is an auditable contract for separating one-model ceilings, union assumptions, scorer choice, and finite-draw sensitivity before interpreting a router-to-oracle gap.

Index Terms—LLM routing, evaluation, stochastic decoding, oracle measurement, coupling, optimizer’s curse, benchmark reliability.

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I. INTRODUCTION

Routing and cascading allocate queries or calls among models in order to trade quality against cost [1, 2]. Their performance is often summarized using a benchmark-specific hindsight-oracle variant. For example, a binary any-correct oracle may credit a query after observing every model’s answer, whereas some score-based variants take a best observed score. Routing benchmarks report oracle references under benchmark-specific conventions; Section II explains how our analysis relates to these references.

That comparison is innocuous when every (query, model) score is a deterministic property. It is different under stochastic decoding. A router selects before seeing the response on which it is scored; the hindsight oracle selects after seeing a collection of responses. Taking a maximum over noisy estimates also creates the familiar optimizer’s curse [3]. Here that maximum estimates the single-commit ceiling. Under the iid and cross-model independence assumptions in Section IV, its upward bias makes the estimated union-minus-ceiling premium downward biased in expectation, not upward biased. This estimation effect is separate from the difference between decision classes. The issue is therefore not simply that an oracle is “too high.” The benchmark has compared two decision rules, with different information and often different numbers of opportunities to succeed.

This paper asks a narrower and testable question:

For a frozen stochastic evaluation protocol, how much does a hindsight union exceed the best success probability obtainable by committing to one model?

Figure 1 makes the resulting information boundary explicit: marginals feed both decision classes, but only the post-response union requires a cross-model coupling.

The answer has two layers. First, the mathematical target must name a coupling across models. Marginal success probabilities alone do not, in general, identify the union probability. We use the product coupling as a transparent reference and keep same-index pairing only as a descriptive sensitivity. Second, any ratio to a router gap must name the router. The numerator $\bar{\Delta}^\perp$ is independent of q , but the denominator $\bar{U}^\perp - \bar{q}$ is not.

Our contributions are:

- 1) **A coupling-aware measurement contract.** We separate a pre-response one-model ceiling from a hindsight union, give the sharp Fréchet bracket allowed by the marginals, and state exactly which additional assumption produces the familiar product formula. Under the stated

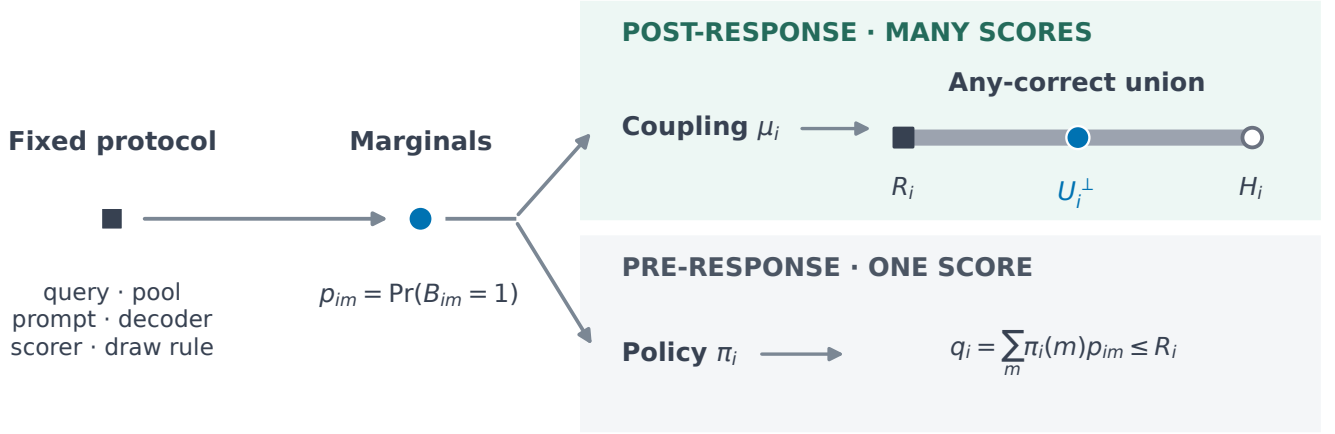


Fig. 1. Schematic measurement contract used throughout the paper (interval position not to scale). The same protocol induces marginal success probabilities, but a hindsight any-correct union also requires a coupling and grants credit from a different decision class. A pre-response one-model policy is instead capped by the querywise clairvoyant ceiling R_i .

conditions, we also decompose a named policy’s product-reference gap into distance to this ceiling and a policy-independent product-union premium, with a matched-protocol reporting recipe.

- 2) **A three-task fixed-tensor audit.** We recompute the same scorer-conditioned functional on GSM8K, MATH-500, and GPQA-Diamond and report the absolute product premium together with its coupling-robust interval and same-index engineering diagnostic.
- 3) **Exhaustive sensitivity rather than a favorable slice.** We report all retained scorer channels, summarize extrema over every one of the 2,047 nonempty model subpools, report every named fixed-pool check, and retain all eight frozen finite- k paths. The complete subpool records are released in machine-readable form.
- 4) **A hash-bound numerical evidence record.** The submitted package contains frozen inputs, the numerical contract, canonical Run-A outputs, and member digests for independent checks of the originally reported numbers; additional revision outputs are recorded separately (Appendix B-E).

Appendix C adds a held-out policy-procedure illustration on the historical MATH-500 archive, not a new routing algorithm or a claim of historically unexposed evaluation data. We do not claim that clairvoyant headroom is attainable, interpret a common draw index as a natural coupling, or aggregate incomparable benchmarks into one headline.

II. RELATED WORK

a) Routing and routing benchmarks: Learned routing and cascading seek cost-aware model selection, from FrugalGPT and Hybrid LLM to RouteLLM [1, 2, 4]. A unified routing-cascading formulation further makes explicit that choosing one model and adaptively invoking another are related but operationally different policies [5]. RouterBench, RouterEval, and LLMRouterBench provide benchmark-specific model pools, metrics, and oracle references [6–8]. Our contribution to

interpreting such references is to separate two questions: what an oracle credits on recorded responses, and what a stochastic any-correct union is identified to be from per-model success probabilities. Proposition 1 gives the sharp interval $[R_i, H_i]$. When that interval is non-degenerate, even exact marginals do not determine a unique union value; eliminating finite-sample estimation bias would not resolve this identification problem. Proposition 2 then relates a named product union to the ceiling for a policy that commits before seeing its response. These are different decision classes even when their numerical values coincide. This provides a way to specify an oracle comparison, rather than a new routing algorithm or a re-evaluation of these benchmarks’ published results.

b) Multi-call inference and repeated sampling: Self-consistency aggregates multiple sampled reasoning paths, compound AI systems study how performance scales with additional model calls, and Mixture-of-Agents combines outputs from several agents [9–11]. Repeated sampling can also expose rare successful responses as inference compute grows [12]. These systems legitimately use more than one response and may inspect intermediate outputs. They are therefore outside our single-commit policy class, not evidence that a one-call router can attain an any-correct union. Our finite- k paths vary the number of archived draws used to estimate a fixed functional; they are not a best-of- k deployment method.

c) Stochastic evaluation and selection bias: LLM scores can vary across repeated executions, even when the nominal protocol is held fixed [13], and benchmark-level variance can have multiple sources [14]. Maximizing noisy quantities is a classical source of upward selection bias [3]. Related responses include decoupling selection from evaluation, simultaneous or winner-focused inference, and general post-selection procedures [15–17]. We do not apply those procedures to manufacture a corrected router score. Instead, we separate the policy classes algebraically and expose scorer, coupling, pool, and finite-draw conditioning in the fixed archive.

d) Recent audits of multi-model ceilings: Our contribution relative to recent audits is a joint measurement result: marginal

success probabilities in general identify an interval rather than a unique hindsight union; every pre-response single-commit policy satisfying the conditions of Proposition 2 is capped by R_i ; and, when the product-coupling premium is positive, its share of the gap changes with the named comparison policy q (Corollary 1). This complements Fowler et al. [18], who distinguish an oracle defined by per-query expected performance from perfect-judge post-hoc selection and develop finite-sample debiasing methods; Garg and Sagtani [19], who study evaluator, truncation, and formatting artifacts; and Chen [20], who study dependence-aware co-failure ceilings. Although $R_i = \max_m p_{im}$, post-hoc union constructions, generic maximum bias, union bounds, and scorer audits are not individually new, the results above make their joint measurement consequences explicit. The repeated historical tensors then audit the product-coupling plug-in premium across retained scorer channels and fixed model pools, while treating finite- k paths as estimator sensitivity rather than a change in the estimand.

Table I pairs these related scopes with the specific results used in this audit.

III. SETUP: THREE DISTINCT OBJECTS

Fix benchmark queries $i = 1, \dots, N$, a model pool $\mathcal{M} = \{1, \dots, M\}$, and the complete prompt, model revision, decoding, and scoring protocol. A fresh response from model m on query i receives a binary score

$$B_{im} \sim \text{Bernoulli}(p_{im}). \quad (1)$$

The parameter p_{im} is conditional on the complete protocol; it is not an intrinsic property of the model or query. Throughout, model pools are nonempty and an empty product equals one. Statements requiring a positive multi-model premium are therefore vacuous for $M = 1$.

The scorer is part of that protocol. Formally these quantities are p_{im}^S for a named scorer S ; we suppress the superscript within one scorer channel and restore it whenever channels are compared. For any query-indexed quantity X_i , write $\bar{X} = N^{-1} \sum_{i=1}^N X_i$.

A. A union requires a coupling

Let μ_i be a joint distribution for $(B_{i1}, \dots, B_{iM})$ having the specified Bernoulli marginals. Its hindsight-union probability is

$$U_i^\mu = \Pr_{\mu_i} \left(\max_m B_{im} = 1 \right). \quad (2)$$

The marginal probabilities do not, in general, identify U_i^μ without a joint protocol. Our primary reference is the specified product coupling,

$$U_i^\perp = 1 - \prod_{m=1}^M (1 - p_{im}). \quad (3)$$

For this historical archive, it is a reference specified from the marginals, not an assertion about the actual cross-model sampling mechanism.

a) *A specified joint sampling protocol*: With the complete protocol fixed and a deterministic scorer applied separately to each response, mutually independent generation randomness across models gives the product coupling. In that case, Eq. (3) is the union probability of the specified experiment; the marginal-identification result is not a reason to discard that value. A hypothetical non-product mechanism is a coordinated categorical sampler: at each decoding step, supply the same uniform random draw to every model, and let each model map it to a token by inverse-CDF sampling under a fixed ordering of its own token support, using draws independent across steps. Each model retains its prescribed marginal token-sampling distribution, while responses across models can become dependent. This requires control of the samplers; shared hardware or a common seed label alone does not specify such a coupling. Nor does this construction generally attain the lower endpoint in Proposition 1. It illustrates a joint sampling choice, not how the historical responses analyzed here were generated.

The second object is the clairvoyant single-commit ceiling

$$R_i = \max_m p_{im}. \quad (4)$$

It assumes knowledge of every p_{im} for this query. It is an upper bound on the expected accuracy of a policy that chooses one model before the scoring response is drawn, not evidence that a feature-based router can attain it.

For compactness, write the sharp marginal upper endpoint as $H_i = \min\{1, \sum_m p_{im}\}$.

Finally, define the product-coupling oracle premium

$$\Delta_i^\perp = U_i^\perp - R_i. \quad (5)$$

This contrast itself crosses decision classes: the union credits success from multiple responses after observing them, whereas the ceiling concerns committing to one model. The same is true of $\bar{U}^\perp - \bar{q}$ for a single-commit policy. Both are well-defined comparisons under matched protocols, but neither is, by itself, an equal-response-budget or attainable routing improvement.

Proposition 1 (Coupling-aware union bracket). *For every joint coupling μ_i with Bernoulli marginals p_{im} ,*

$$R_i \leq U_i^\mu \leq H_i. \quad (6)$$

Moreover, the set of attainable values of U_i^μ is exactly this interval. Consequently $\Delta_i^\perp \geq 0$.

The bracket cannot in general be narrowed from the marginals alone. For $p_{i1} = p_{i2} = 1/2$, identical success events attain the lower endpoint $1/2$, whereas disjoint success events attain the upper endpoint 1.

If $m^* \in \arg \max_m p_{im}$ and $p_i^* = p_{im^*}$, the product premium also has the exact factorization

$$\Delta_i^\perp = (1 - p_i^*) \left[1 - \prod_{m \neq m^*} (1 - p_{im}) \right]. \quad (7)$$

Under the product reference, the premium is therefore the probability that a best model fails multiplied by the probability that at least one alternative succeeds. This interpretation does not extend automatically to other couplings.

TABLE I
SCOPE RELATIVE TO ROUTING BENCHMARKS AND THE CLOSEST CONTEMPORANEOUS AUDITS. OVERLAP IN INDIVIDUAL INGREDIENTS IS ACKNOWLEDGED; THE LAST COLUMN STATES A CONCRETE RESULT ESTABLISHED HERE RATHER THAN A PRIORITY CLAIM.

Work	Relevant scope	Concrete result established here
RouterBench, RouterEval, and LLMRouterBench [6–8]	Routing evaluation, including benchmark-specific model pools, metrics, and oracle references	Proposition 1 gives the sharp set of stochastic any-correct unions compatible with fixed marginals, while Proposition 2 separates that post-response object from the pre-response single-commit ceiling.
Fowler et al.	Expected-performance oracle, post-hoc selection, and finite-sample debiasing	Proposition 1 gives the sharp interval $[R_i, H_i]$. If non-degenerate, it remains non-degenerate even with exact marginals, separately from estimation bias.
Garg–Sagtani	Evaluator, truncation, and formatting artifacts in reported ceilings	For a fixed scorer, Proposition 1 characterizes the attainable union probabilities over couplings with the same marginals.
Chen	Dependence-aware all-model co-failure ceilings	Proposition 2 separates $\bar{U}^\perp - \bar{q}$ into $\bar{R} - \bar{q}$ and $\bar{\Delta}^\perp$; Corollary 1 makes the corresponding fraction policy-dependent.

Figure 2 illustrates the size of this distinction in a homogeneous toy case and separates the interval itself from the pool-size trend under the named product coupling.

The statement requires no independence. Independence enters only when the specific value in Eq. (3) is chosen.

B. A named single-commit policy

Let a policy choose $M_i \sim \pi_i$ using information available before the scoring response, and assume that choice is conditionally independent of that fresh response. Its expected accuracy on query i is

$$q_i = \sum_m \pi_i(m) p_{im}. \quad (8)$$

Proposition 2 (Policy-class ceiling and exact decomposition). *Every such one-model policy satisfies $q_i \leq R_i$. Averaging over the fixed benchmark,*

$$\bar{U}^\perp - \bar{q} = \underbrace{(\bar{R} - \bar{q})}_{\text{clairvoyant potential headroom}} + \underbrace{\bar{\Delta}^\perp}_{\text{product-coupling union premium}}. \quad (9)$$

Both terms are non-negative, but only the second is independent of the named comparison policy.

This scope excludes cascades, parallel calls, ensembles, and policies allowed to inspect the response before selecting it.

For $\bar{U}^\perp > \bar{q}$, define the diagnostic share

$$s(q) = \frac{\bar{\Delta}^\perp}{\bar{U}^\perp - \bar{q}}. \quad (10)$$

Corollary 1 (The gap fraction is policy-dependent). *If $\bar{\Delta}^\perp > 0$ and $\bar{q}_1 < \bar{q}_2 \leq \bar{R}$, then $s(q_1) < s(q_2)$. Therefore a percentage of “the routing gap” is undefined until q , its training split, and its evaluation protocol are named.*

Within this policy class, $\bar{q} \leq \bar{R}$. Hence, when $\bar{\Delta}^\perp > 0$, $\bar{U}^\perp - \bar{q} = \bar{\Delta}^\perp + (\bar{R} - \bar{q}) \geq \bar{\Delta}^\perp$ and therefore $0 < s(q) \leq 1$, with equality only when $\bar{q} = \bar{R}$. Under the stated preconditions $s(q)$ cannot be negative; $s(q) > 1$ requires $\bar{q} > \bar{R}$ and therefore a comparison outside the pre-response single-commit policy ceiling.

IV. FINITE ARCHIVED SAMPLES

The population notation above clarifies the decision classes. The actual release, however, begins from a finite archived tensor. For scorer S , let $B_{imj}^S \in \{0, 1\}$ be the stored score of draw j . The mathematical subscripts are ordered query, model, draw; the serialized tensor axes described below are ordered model, query, draw. For a fixed draw-index set J with $|J| = k$, define

$$\hat{p}_{im}^S(J) = \frac{1}{k} \sum_{j \in J} B_{imj}^S, \quad (11)$$

$$\hat{U}_{i,A}^{S,\perp}(J) = 1 - \prod_{m \in A} \{1 - \hat{p}_{im}^S(J)\},$$

$$\hat{R}_{i,A}^S(J) = \max_{m \in A} \hat{p}_{im}^S(J), \quad (12)$$

$$\hat{\Delta}_{i,A}^{S,\perp}(J) = \hat{U}_{i,A}^{S,\perp}(J) - \hat{R}_{i,A}^S(J).$$

All headline values first apply these nonlinear operations within a query and then average with equal query weights. Averaging the cell probabilities before taking a maximum or product generally gives a different estimand. For example, two queries with success vectors $\mathbf{p}_1 = (1, 0)$ and $\mathbf{p}_2 = (0, 1)$ have mean querywise ceiling one, whereas the maximum after averaging the model columns is one half.

Equation (11) also has an exact fixed-tensor interpretation. For each model independently, draw an index uniformly from J and ask whether at least one selected archived response is correct. Its conditional probability, given the tensor, is $\hat{U}_{i,A}^{S,\perp}(J)$. Product is therefore the exact conditional union probability for this specified empirical resampling experiment. Independence is imposed by that rule, not inferred from the historical generation schedule.

Proposition 3 (Structural finite- k facts). *For every binary archived tensor and nonempty pool A : (i) at $k = 1$, $\hat{U}_{i,A}^{S,\perp} = \hat{R}_{i,A}^S$ and the plug-in premium is zero; (ii) when J contains all archived draws, every permutation gives identical metrics; and (iii) neither the premium nor its components must be monotone along nested prefixes.*

The third point matters empirically: a rising path should not be called a theorem of convergence. Our finite- k experiment uses eight frozen, outcome-independent permutations and

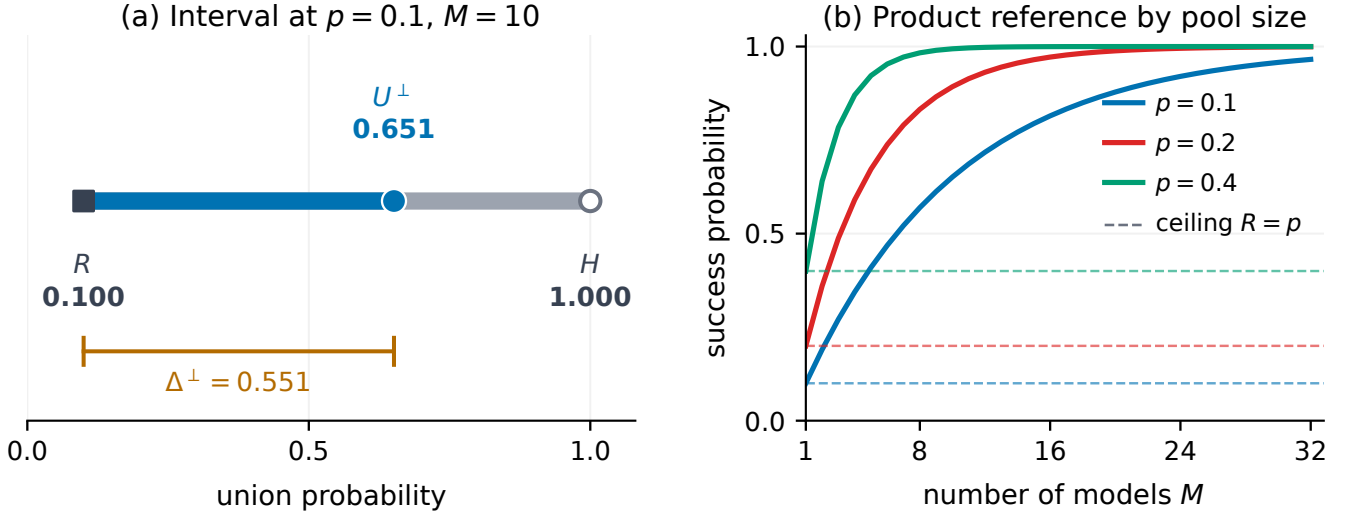


Fig. 2. Toy example with ten homogeneous marginals $p_m = 0.1$, the single-commit ceiling is 0.1, whereas the named product coupling gives union $1 - 0.9^{10} = 0.651$ and premium 0.551. The right-hand curve varies model count under this coupling; other couplings can attain any value in the Fréchet interval.

retains every path. Their construction is SHA-256-derived and does not use response outcomes. The summaries are descriptive order sensitivity, not independent replications, confidence limits, or deployment call-budget curves.

Under additional iid within-cell and cross-model independent-stream assumptions, the product plug-in is unbiased, whereas the plug-in maximum has nonnegative bias, possibly zero, bounded by $\sqrt{\log(M)/(2k)}$. Consequently, under those assumptions the plug-in premium has non-positive bias: it cannot systematically inflate the premium in expectation. This sign concerns repeated-sample expectation, not the error direction of this one archived realization. At the realized full-pool values $M = 11$ and $k = 30$, the worst-case bias-magnitude bound is 0.1999, or 19.99 percentage points, and is too loose to be a useful data-specific correction. We give the conditional result in Appendix A-F. The archived tensor does not verify its population assumptions, so we apply no model-based correction.

V. RETROSPECTIVE AUDIT PROTOCOL

A. Design, queries, and model pool

This is a historical-pilot-informed retrospective fixed-tensor audit. The benchmark queries, model pool, scorer channels, and archived responses are treated as fixed cases; no query-, model-, or benchmark-population inference is made.

The tensor contains 30 stored responses for every selected query-model cell from 11 open-weight instruction models. The generation logs record temperature 0.2 and top- p 1.0. We analyze 500 GSM8K queries [21], selected uniformly without replacement from the 1,319 test items with NumPy `default_rng(42)` and then sorted by source index; an independent replay exactly reproduces every archived query ID. We also analyze the fixed 500-problem MATH test subset released with *Let's Verify Step by Step* [22–24], and all 198 GPQA-Diamond questions [25]. The canonical tensor order is

model, query, draw, giving 165,000 binary scores per scorer channel for each math task and 65,340 for GPQA-Diamond.

The fixed 11-model pool spans Mistral, DeepSeek-Distill-Qwen, three Qwen2.5 sizes, Phi, OLMo, Yi, Granite, Gemma, and Llama variants. Exact identifiers are listed in Table XI. Historical repository and tokenizer revisions were not all recorded, so the results characterize archived responses rather than current repositories.

Table II records the tensor dimensions and every retained scorer channel used by the audit.

B. Scorer channels and leakage boundary

Changing a scorer changes $B_{im,j}^S$ and therefore every downstream marginal, ceiling, union, and premium. GSM8K and GPQA-Diamond each retain legacy, v1, and v2 channels; MATH-500 retains a legacy lexical channel and the `math500-mathverify` channel backed by Math-Verify 0.9.0 [26]. Every retained channel is computed on the same query, model, draw, and raw-response identities within a benchmark, and all are reported. These channels are alternative deterministic scoring policies applied to the same archived responses, not independent evaluator replications. Their min-max span is therefore scorer sensitivity, not statistical uncertainty.

The GSM8K and GPQA v2 policies were developed after inspecting limited real outputs. They remain labeled post-output-developed, not outcome-blind; the later human check below does not change that history. MATH-500's policy uses a bounded boxed-answer extractor and a frozen Math-Verify configuration, but the release reruns score-tensor-to-result analysis rather than re-executing raw-text scoring. The display channels are presentation labels, not privileged ground truth or a claim of outcome-blind channel selection.

The retained working records distinguish scorer adoption, premium computation, and display designation. On 29 July 2026, the decision to rescore MATH-500 with Math-Verify and

TABLE II

FROZEN AUDIT MATRIX. ALL SCORER CHANNELS WITHIN A BENCHMARK USE THE SAME QUERY, MODEL, DRAW, AND RAW-RESPONSE IDENTITIES; THEY ARE ALTERNATIVE DETERMINISTIC POLICIES, NOT INDEPENDENT EVALUATOR REPLICATIONS.

Benchmark	N	Tensor shape	Display scorer (status)	Additional channels
GSM8K	500	$11 \times 500 \times 30$	semantic v2 (post-output-developed)	marker-exact v1; legacy lexical
MATH-500	500	$11 \times 500 \times 30$	Math-Verify (audited candidate)	legacy lexical
GPQA-Diamond	198	$11 \times 198 \times 30$	endorsed-choice v2 (post-output-developed)	final-choice v1; legacy lexical

official gold preceded the recovered premium computation; the explicit instruction to use that tensor throughout the MATH tables, figures, and text followed it. On 1 August, the table-building code designated GSM8K and GPQA v2 before the earliest corresponding final-v2 premium computations recoverable from these records. Output inspection and scoring diagnostics had already informed their development. The 8 August release contract formalized all three display choices after the premiums were available. These recovered events do not establish an outcome-blind initial display freeze or rule out earlier unrecorded computations. Every retained channel is reported in Table V. Scorer disagreement is reported as evaluator conditioning; it is not automatically annotation error.

a) Post-review human check: Three contributors provided two initial judgments per archived response for 60 distinct cells: 30 GSM8K and 30 GPQA-Diamond, with 15 from each frozen-v2 score side per benchmark. The task was to judge the response’s final answer against the supplied reference, not to independently re-solve the benchmark. The packets withheld automated scores, model-identifying metadata, and other contributors’ judgments. Only matching, definite initial judgments were treated as an analysis consensus. GSM8K had such a consensus on 26 of 30 cells, all 26 agreeing with frozen v2; GPQA-Diamond had a consensus on 11 of 30, all 11 agreeing. The remaining 4 and 19 cells were retained as undetermined, not excluded from the sample. Thus the check supplies limited evidence on the determined subsets, with substantially less coverage on GPQA-Diamond; it does not validate the full archive or other scorer channels. Appendix D reports the sampling, initial disagreements, and the decision to forgo planned follow-up after seeing the initial-judgment summary but before comparison with automated scores. No scorer or original audit result was changed.

C. Frozen analysis plan

For every benchmark and scorer we compute the full-pool empirical product union, empirical one-model ceiling, absolute premium, Fréchet endpoints, and same-index paired union. We then enumerate all $2^{11} - 1 = 2,047$ nonempty subpools in canonical bitmask order. Singleton premiums must be exactly zero. The release also evaluates fixed provenance-group pools and every leave-one-declared-group-out pool. These are sensitivity checks over a fixed set, not iid draws from a model population.

For finite- k sensitivity, $k \in \{1, 2, 5, 10, 15, 20, 25, 30\}$. Each of eight SHA-256-derived, outcome-independent permutations supplies nested draw prefixes, and the same permutation is

used across scorer channels within a benchmark. The eight permutations are enumerated per benchmark in the released numerical contract, and all paths are retained. At $k = 30$, every prefix is the full archive and must reproduce the headline exactly.

Figure 3 summarizes this frozen-tensor path from archived binary scores to the querywise functionals and hash-bound outputs.

The original Run-A audit contains no empirical named-policy accuracy or router-dependent ratio. The separate revision experiment in Appendix C adds such a policy on a 300/100/100 train/tune/evaluation split of the MATH-500 archive. Its policy fitting, tuning, and prediction stages exclude evaluation scores; the author’s prior exposure to the archive is not erased by this split. This illustration does not alter the original audit’s inputs or results.

D. Released inputs and numerical checks

The submitted supplement contains the frozen score tensors, their ordered model/query/draw identities, the numerical contract, canonical Run-A outputs, and hash records. These support checks of the score-tensor-to-result calculations. The original Run-A checks cover eleven declared scientific invariants, including complete subpool enumeration, full-draw identity, scorer completeness, absence of named-policy outputs, and absence of cross-benchmark aggregation. The absence-of-policy-output invariant describes that original artifact only; the revision’s policy outputs have a separate evidence chain.

The bundle contains an analysis receipt and member digests, but not the analysis source or referenced release-control files, including `release/source_manifest_v1.json`, `release/anonymous_allowlist_v1.json`, `configs/reviewed_source_paths_v1.json`, and `release/release_policy_v1.json`. It therefore does not supply a reusable implementation for new tensors or the unshared internal execution procedures. Appendix B-E distinguishes the submitted bundle from the descriptive and policy outputs added in this revision.

VI. RESULTS

a) Scope of the numerical results: All reported quantities describe fixed archived scores under the named scorer, pool, and coupling, with equal query weights within each benchmark. Empirical identification intervals range over couplings compatible with the empirical marginals; scorer, pool, and prefix ranges describe their respective enumerated sensitivity sets, not sampling uncertainty. These quantities are not population

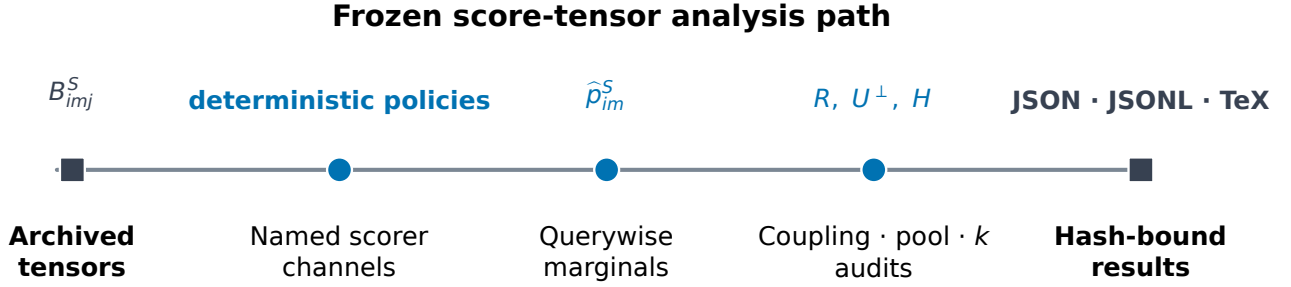


Fig. 3. Score-tensor-to-paper path. Nonlinear functionals are computed querywise before equal-weight averaging within each benchmark; the result files retain the numerical outputs and their input bindings.

guarantees; the main audit does not measure a router’s performance. The separate policy evaluation is reported in Appendix C. Section VIII states the shared limits; the text and appendices below retain qualifications specific to each comparison.

Figure 4 places the product reference and same-index diagnostic inside the marginally identified interval for each benchmark.

A. Where the product-coupled premium occurs

The display-channel full-pool premiums are 0.371 percentage points on GSM8K, 3.542 on MATH-500, and 5.100 on GPQA-Diamond (Table III). To locate these averages within each benchmark, we added a querywise analysis in response to review, after seeing the original results. It uses the same frozen scores, all 11 models, all 30 archived draws per cell, and every retained scorer channel; the existing display designations are unchanged.

Fix a benchmark and scorer, and suppress the scorer, full-pool and full-draw indices in Eq. (11). Equation (7), applied to the empirical marginals, gives

$$\hat{\Delta}_i^\perp > 0 \iff \max_m \hat{p}_{im} < 1 \text{ and } \#\{m : \hat{p}_{im} > 0\} \geq 2.$$

Let n_+ count these queries. For $n_+ > 0$, define their conditional mean $\hat{d}_+ = n_+^{-1} \sum_{i: \hat{\Delta}_i^\perp > 0} \hat{\Delta}_i^\perp$. Then, with equal query weights within each benchmark,

$$\hat{\Delta}^\perp = \frac{n_+}{N} \hat{d}_+.$$

Thus the average separates exactly into the fraction of queries with a positive product premium and its mean size on those queries. If $n_+ = 0$, the overall premium is zero and the conditional mean is undefined.

On the display channels, positive premiums occur on 13 of 500 GSM8K queries, 113 of 500 MATH-500 queries, and 73 of 198 GPQA-Diamond queries. Their conditional means are 14.255, 15.672, and 13.834 pp, respectively (Table IV). For GPQA-Diamond relative to GSM8K on these display channels, the positive-query fraction is 14.18 times as large, whereas the conditional mean is 0.9705 times as large, computed from the unrounded summaries. Thus the larger average premium in

this comparison reflects a larger fraction of positive-premium queries, not a larger premium on those queries. This is a factorization of the fixed-archive averages, not a causal explanation of benchmark differences.

The zero-premium queries fall into three mutually exclusive cases: at least one model has all 30 archived scores correct; all model counts are zero; or exactly one model has a positive count, and that count is smaller than 30. In that order, the display-channel counts are (485, 0, 2) for GSM8K, (372, 10, 5) for MATH-500, and (117, 5, 3) for GPQA-Diamond.

These cases also characterize collapse of the empirical Fréchet bracket. Write $\hat{H}_i = \min\{1, \sum_m \hat{p}_{im}\}$. If $\hat{R}_i = 1$, then $\hat{U}_i^\perp = \hat{H}_i = 1$. Otherwise, Eq. (7) makes a zero product premium equivalent to all marginals other than a chosen maximizer being zero, so $\hat{H}_i = \hat{R}_i$. Conversely, the bracket in Proposition 1 gives zero premium whenever $\hat{H}_i = \hat{R}_i$. Hence

$$\hat{\Delta}_i^\perp = 0 \iff \hat{H}_i = \hat{R}_i.$$

The interval constructed from the empirical marginals therefore collapses on every zero-premium query, so every coupling with these empirical marginals has the same union value. This conclusion concerns the frozen scores: archived counts of 30/30 or 0/30 do not establish population success probabilities of one or zero, or collapse of the corresponding population bracket.

For GSM8K, the mean querywise plug-in ceiling is $\hat{R} = 98.980\%$. The largest in-sample single-model mean on those same scores is 96.320%. These are two aggregation orders, not measured performance of a learned router. Appendix B-B reports their difference for each benchmark and scorer.

B. Empirical coupling bounds and the product reference

For the empirical marginals, the sharp average-premium intervals over compatible couplings are $[0, 0.473]$, $[0, 4.787]$, and $[0, 7.744]$ percentage points for GSM8K, MATH-500, and GPQA-Diamond. The zero lower endpoint follows from the nested-event construction in the proof of Proposition 1; the data determine the upper endpoints. The positive product values select one specified reference within these intervals.

If independent sampling is part of the complete protocol, product is the correct joint reference for that experiment. The intervals answer a different question: what the marginals alone

TABLE III

FULL-POOL DISPLAY-CHANNEL RESULTS. SCORE LEVELS ARE PERCENTAGES; PREMIUMS AND DIFFERENCES ARE PERCENTAGE POINTS (PP). PRODUCT DENOTES THE SPECIFIED COUPLING REFERENCE; FRÉCHET INTERVALS USE EMPIRICAL MARGINALS.

Benchmark	$\hat{R}$	$\hat{U}^\perp$	$\hat{\Delta}^\perp$	Fréchet premium	Paired — product
GSM8K	98.980	99.351	0.371	[0, 0.473]	−0.011
MATH-500	87.713	91.255	3.542	[0, 4.787]	−0.129
GPQA-Diamond	82.694	87.794	5.100	[0, 7.744]	−0.269

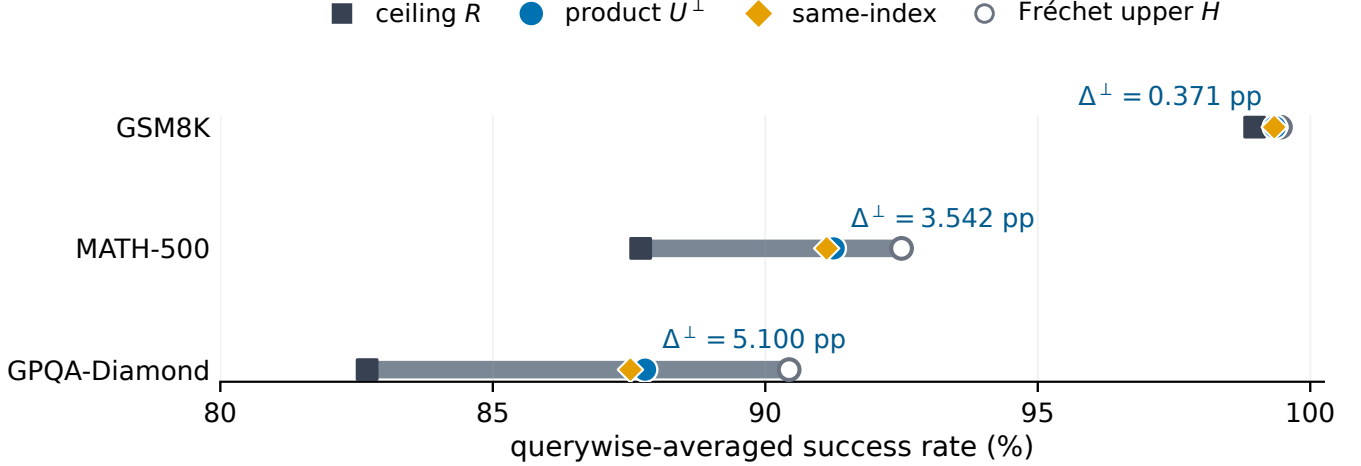


Fig. 4. Full-pool display-channel identification view (%). Gray segments join the empirical querywise-averaged Fréchet endpoints $[\hat{R}, \hat{H}]$ (abbreviated in the legend), not confidence limits. Blue points mark product references; orange diamonds mark same-index diagnostics.

identify when a joint protocol is not specified. The coordinated sampler in Section III illustrates how marginal sampling can be preserved while changing cross-model dependence; it is not an account of this archive. The empirical bounds do not show that an actual deployment has zero premium or that the product premium is an attainable one-call gain.

The archived same-index union lies 0.011, 0.129, and 0.269 points below the product value, respectively. These average union values are close to product in the observed engineering pairing, rather than near the zero-premium endpoint. This is positive descriptive evidence about that pairing, not evidence that the unknown deployment coupling has zero premium. Nor does average proximity test independence or establish querywise agreement: same-index pairing is one empirical coupling, and its difference from product can have either sign (Appendix B-A).

C. The qualitative ordering survives scorer choice

Across retained channels, premiums range from 0.359 to 0.398 points for GSM8K, 3.542 to 3.754 for MATH-500, and 5.100 to 6.632 for GPQA-Diamond (Table V). Thus every retained scorer preserves the qualitative ordering of these three fixed case studies, but GPQA-Diamond has a substantially wider scorer range. We do not average channels or benchmarks, and agreement in ordering does not validate any scorer.

Figure 5 displays every retained deterministic scorer channel rather than selecting only the designated display values.

D. Pool composition is a material conditioning choice

Across the 12 named pool analyses on each display channel—the full pool, two named fixed subsets, and nine leave-one-declared-group-out pools—the absolute product premium ranges from 0.326 to 0.567 percentage points on GSM8K, from 2.508 to 3.837 points on MATH-500, and from 0.550 to 7.729 points on GPQA-Diamond (Table VI). The corresponding full-pool values are 0.371, 3.542, and 5.100 points. These frozen fixed-tensor contrasts are descriptive sensitivity analyses, not optimized deployment policies or draws from a model population.

Figure 6 compares the three named fixed pools and uses both color and hatching to distinguish them.

E. Finite-draw paths diagnose plug-in estimator sensitivity

Here $k = |J|$ controls how many archived draws enter $\hat{p}_{im}^S(J)$ and its derived plug-in quantities. It changes the estimator, not the population estimand in Section III.

Tables VII and XIII give two descriptive checks. Comparing intermediate means and ranges with the full-archive endpoint shows sensitivity to retaining only a prefix of the draws. At a fixed k , the min–max range shows sensitivity to ordering across the eight specified paths. The full-archive endpoint is a reference for these comparisons, not the known population value.

Proposition 3 explains the endpoints: binary empirical marginals force a zero premium at $k = 1$, and all eight paths

TABLE IV

QUERYWISE PRODUCT-PREMIUM SUMMARIES FOR EVERY RETAINED SCORER. A DAGGER MARKS EACH BENCHMARK’S EXISTING DISPLAY CHANNEL. THE CONDITIONAL MEAN USES ONLY POSITIVE-PREMIUM QUERIES; THE OVERALL MEAN USES ALL N QUERIES. BOTH ARE IN PP, ROUNDED DIRECTLY FROM EXACT VALUES.

Scorer	Positive queries n_+/N	Conditional mean $\hat{d}_+$ (pp)	Overall mean $\hat{\Delta}^\perp$ (pp)
GSM8K			
gsm8k_v2 [†]	13/500	14.255	0.371
gsm8k_v1	17/500	10.549	0.359
gsm8k_legacy	14/500	14.207	0.398
MATH-500			
math500_mathverify [†]	113/500	15.672	3.542
math500_legacy	120/500	15.643	3.754
GPQA-Diamond			
gpqa_v2 [†]	73/198	13.834	5.100
gpqa_v1	79/198	16.621	6.632
gpqa_legacy	79/198	14.172	5.654

Rounding these printed values again can differ from rounding the exact values directly.

TABLE V

FULL-POOL PRODUCT PREMIUMS (PP) FOR EVERY RETAINED SCORER ON THE SAME ARCHIVED RESPONSES. DISPLAY DESIGNATIONS ARE PRESENTATION LABELS, NOT A CLAIM OF OUTCOME-BLIND SELECTION OR INDEPENDENT SCORER VALIDATION.

Benchmark	Channel	Role	Premium
GSM8K	gsm8k_v2	audited candidate, post-output developed	0.371
	gsm8k_v1	sensitivity	0.359
	gsm8k_legacy	sensitivity	0.398
MATH-500	math500_mathverify	audited candidate	3.542
	math500_legacy	sensitivity	3.754
GPQA-Diamond	gpqa_v2	audited candidate, post-output developed	5.100
	gpqa_v1	sensitivity	6.632
	gpqa_legacy	sensitivity	5.654

TABLE VI

DISPLAY-CHANNEL PRODUCT PREMIUMS (PP) FOR NAMED FIXED POOLS. THE LEAVE-ONE-GROUP-OUT ROW IS THE MINIMUM–MAXIMUM ACROSS NINE DECLARED GROUP REMOVALS, DESCRIBING A FIXED SENSITIVITY SET.

Pool analysis	GSM8K	MATH-500	GPQA-Diamond
Full 11-model pool	0.371	3.542	5.100
One per declared group	0.378	3.837	6.526
Qwen2.5 size sweep	0.390	2.508	0.550
Leave-one-group-out range	0.326–0.567	3.311–3.648	4.408–7.729

Leave-out minima/maxima are Mistral/Phi for GSM8K, Granite/Phi for MATH-500, and Granite/Qwen2.5 for GPQA-Diamond.

contain the same complete archive at $k = 30$. The first is not evidence of a zero population premium; the second is agreement by design, not evidence of convergence. Individual paths need not be monotone. Neither intermediate values nor their spread provide independent replications, confidence intervals, a deployment call-budget curve, or a stopping rule for collecting responses (Section VIII). Figure 7 displays the structural endpoints and intermediate order sensitivity together.

F. The full pool need not maximize the premium

Adding models makes the product union monotone, but not the premium. The largest observed premiums occur at pool sizes 3, 5, and 6 (Table VIII). As maxima over 2,047 correlated fixed-tensor values, they are descriptive and

potentially selection-optimistic, not deployment policies or model-population estimates.

Figure 8 shows the exact min–max envelope, median, and mean at each pool size over all enumerated nonempty subpools.

G. What the released artifacts let a reader check

The canonical Run-A results report PASS for all eleven declared scientific invariants. The submitted tensors, numerical contract, and machine-readable outputs allow independent number checks, while the in-bundle manifest records member hashes. This evidence starts at frozen scores: it does not validate the scorers or regenerate historical responses. The added querywise summaries and aggregation-order comparison

Product-coupling premium under every retained scorer policy

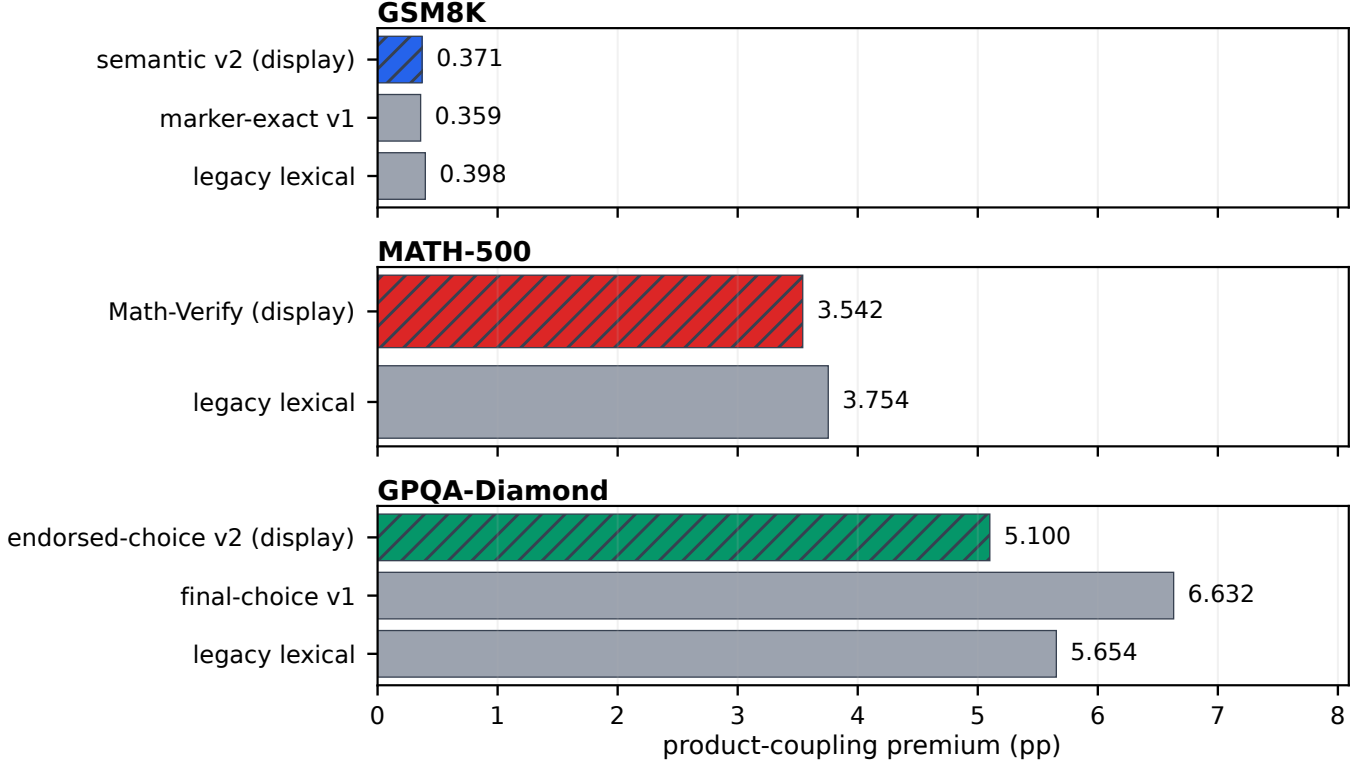


Fig. 5. Full-pool product premiums (pp) under all retained scorers. Hatched colored bars mark display channels; gray bars mark other channels on the same responses. Their spread describes scorer sensitivity.

TABLE VII

MEAN DISPLAY-CHANNEL PREMIUM ACROSS EIGHT FIXED NESTED DRAW PERMUTATIONS (PP). INTERMEDIATE MEANS DIAGNOSE PREFIX SENSITIVITY. HERE k CHANGES THE ESTIMATOR, NOT THE ESTIMAND: THE $k = 1$ ZERO IS STRUCTURAL, AND AGREEMENT AT $k = 30$ FOLLOWS FROM USING THE SAME COMPLETE ARCHIVE, NOT CONVERGENCE.

Benchmark	$k = 1$	$k = 5$	$k = 10$	$k = 20$	$k = 30$
GSM8K	0.000	0.236	0.317	0.349	0.371
MATH-500	0.000	2.372	3.024	3.398	3.542
GPQA-Diamond	0.000	3.559	4.398	4.757	5.100

use the same scores; their separate result record, and the policy experiment’s distinct chain, are identified in Appendix B-E.

VII. DISCUSSION

a) What the audit establishes: The results connect three questions that an oracle comparison needs to keep separate. Proposition 1 gives the sharp set of any-correct union probabilities compatible with the marginals: $[R_i, H_i]$. When this interval is non-degenerate, even exact marginal success probabilities leave the union value unidentified. Proposition 2 separates the product-union gap into a policy-dependent distance from the single-commit ceiling and a product-coupling premium. The fixed-archive audit then shows how the reported premium depends on the declared scorer, coupling, and pool, while finite- k paths describe sensitivity of its plug-in estimator.

b) Why this matters: An oracle reference can be useful without being a deployable routing target. If a distance to an any-correct reference is read as routing opportunity,

the interpretation needs a named coupling, a decision class and response budget, a pool, and a scorer. The coupling issue is a property of the functional in Proposition 1, not a diagnosis of a particular benchmark’s implementation. Across couplings compatible with the same marginals, zero premium is admissible; this does not establish that the premium under a specified joint sampling protocol is zero. Our own product-union premium is also a cross-decision-class contrast, not an exception to this reporting requirement. Under a specified independent joint protocol, product is the corresponding union probability; its positive premium alone still does not establish an attainable one-call gain.

c) Estimation bias is a separate question: Under the iid within-cell and cross-model independence assumptions of Appendix A-F, the product plug-in is unbiased and the ceiling plug-in has nonnegative bias. The premium is therefore conservative in expectation under those assumptions: its bias is non-positive. This explains how the optimizer’s curse mentioned

Named model-pool sensitivity on the fixed tensors

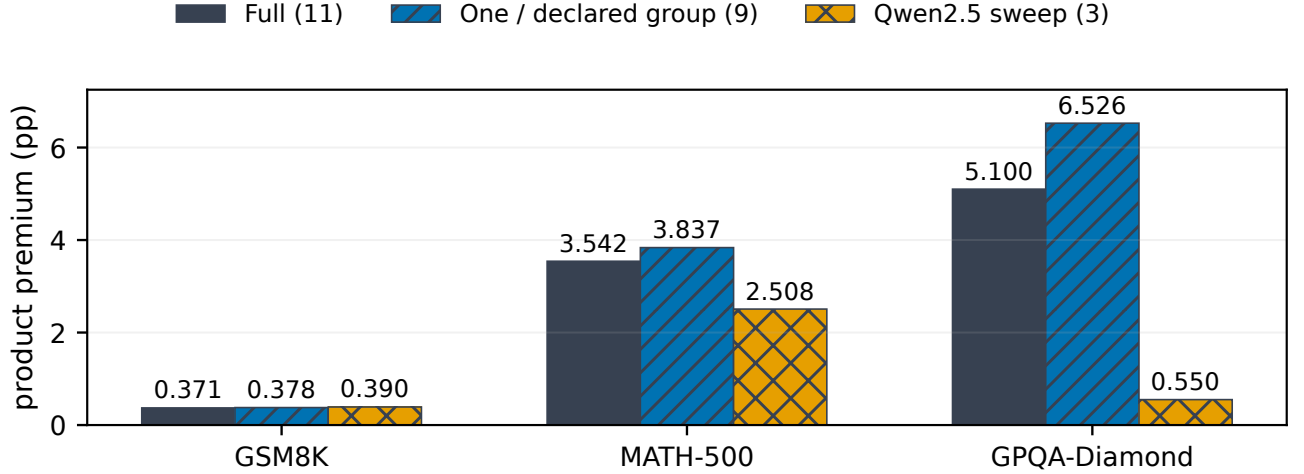


Fig. 6. Product premiums (pp) for three named fixed pools. The Qwen2.5 sweep changes both composition and size (three versus eleven models), so the contrast does not isolate either effect. The supplement records fixed-pool results for every retained scorer.

TABLE VIII
DISPLAY-CHANNEL PRODUCT PREMIUMS OVER ALL $2^{11} - 1 = 2,047$ NONEMPTY SUBPOOLS. MAXIMA ARE DESCRIPTIVE, SELECTION-OPTIMISTIC EXTREMA OF THIS ENUMERATED SET.

Benchmark	Full 11-model premium	Largest subpool premium	Pool size at largest
GSM8K	0.371 pp	2.682 pp	3
MATH-500	3.542 pp	6.859 pp	5
GPQA-Diamond	5.100 pp	9.627 pp	6

in the Introduction acts here; it does not inflate the premium. It also does not make a realized archived premium a guaranteed lower bound, certify the archive’s sampling assumptions, or remove the need to specify the comparison’s decision classes.

d) A recipe for evaluating a reader’s router: Appendix C illustrates this calculation for a policy evaluated on a held-out role within the historical archive, with the historical exposure limitation stated separately. The general recipe below specifies the requirements for a reader’s independently evaluated policy. For a reader’s router, Eq. (9) separates its distance from the single-commit ceiling from the additional product-union premium. First fix the policy, model pool, scorer, and decoding protocol. Evaluation queries and outcomes must be excluded from training, tuning, and policy selection; scorer development and model-pool selection must also be disclosed. Freezing a policy after inspecting those outcomes does not turn them into a holdout. Record π_i , its distribution over the declared models, using only information available before the scored response. A deterministic router gives a point mass. The conditional independence requirement in Proposition 2 still applies.

Obtain repeated evaluation responses under the common protocol and compute $\hat{p}_{im}$ as in Eq. (11). A full-pool comparison requires outcomes for every query–model cell, not just the models the router selected. With the paper’s equal query

weights, estimate the policy’s expected accuracy by

$$\hat{q} = \frac{1}{N} \sum_i \sum_m \pi_i(m) \hat{p}_{im}.$$

Use those same marginals, queries, pool, scorer, and weights to compute $\hat{R}$, $\hat{U}^\perp$, and $\hat{\Delta}^\perp = \hat{U}^\perp - \hat{R}$. Apply the maximum and product within each query before averaging (Section IV). The product union is the conditional union probability for independent per-model resampling from these empirical marginals, not a certification of the historical cross-model coupling. These plug-in quantities are defined for $k \geq 1$ draws. At $k = 1$ the premium is forced to zero (Proposition 3); it is also zero for a one-model pool. $k > 1$ and a pool with at least two models avoid those forced zeros but guarantee neither a positive premium nor accurate estimation.

If $\hat{U}^\perp > \hat{q}$, form

$$\hat{s} = \frac{\hat{\Delta}^\perp}{\hat{U}^\perp - \hat{q}}.$$

Because π_i is a probability distribution on that pool, the matched calculation satisfies $\hat{q} \leq \hat{R} \leq \hat{U}^\perp$ algebraically. Thus $\hat{s} \in [0, 1]$ when its denominator is positive. This is not a finite-sample accuracy guarantee or a test for leakage: the inequality

Eight finite-draw paths share the full-archive endpoint

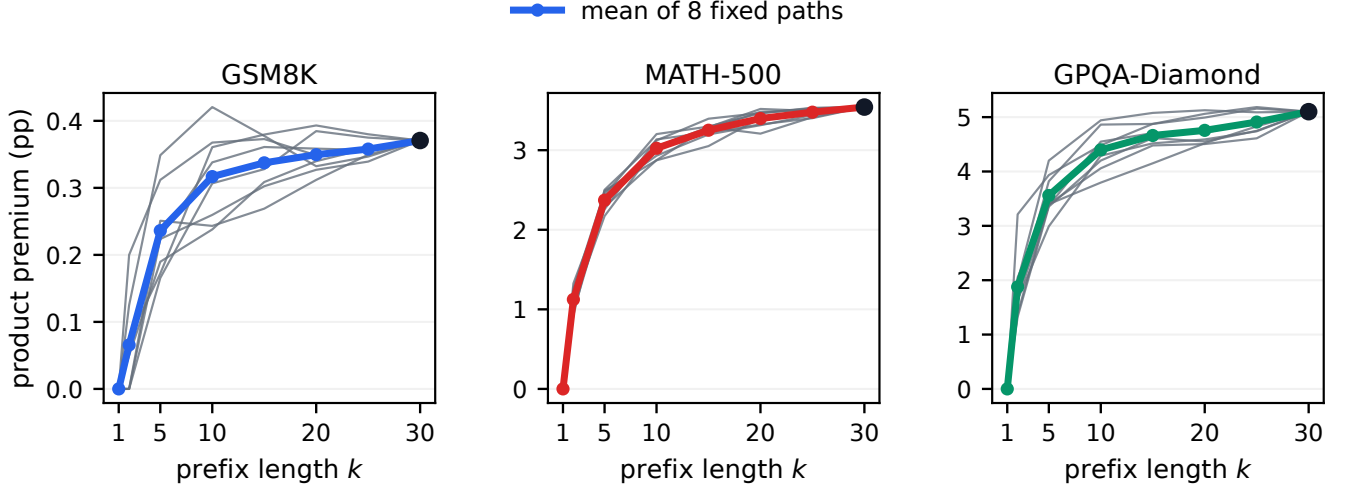


Fig. 7. Product premiums (pp) along eight fixed nested-prefix paths, with their mean in bold. The $k = 1$ zero and shared $k = 30$ value are structural; intermediate dispersion diagnoses order sensitivity, not convergence or uncertainty coverage.

can hold even for a policy selected using evaluation outcomes. The holdout requirement must be checked from the data and selection history.

For the exact quantities, Eq. (10) is defined only when $\bar{U}^\perp > \bar{q}$. Within the stated policy class, $0 < s(q) < 1$ means both terms in Eq. (9) are positive, while $s(q) = 1$ means $\bar{q} = \bar{R}$ and a positive premium. When $\bar{\Delta}^\perp > 0$, Corollary 1 shows that the share rises as the comparator approaches $\bar{R}$; it is not a policy-independent measure of routing opportunity. If $\bar{\Delta}^\perp = 0$ and the denominator is positive, the share is zero. A zero denominator makes the share undefined; a negative one cannot occur under these exact preconditions. The term $\bar{R} - \bar{q}$ is not proved learnable, and the union premium is not an equal-cost routing gain.

An exact $s(q) > 1$ is incompatible with those preconditions. Published finite-sample estimates need more care: accuracy measured on separate realized routed responses need not obey the matched-marginal inequality. A share above one can then reflect sampling disagreement or a mismatch in policy class, protocol, split, scorer, or response reuse. It is a reason to audit the comparison, not proof from the numbers alone that a particular condition failed. A value above one from the matched formula instead indicates an invalid input, implementation, or reporting step. An empirical report should distinguish the policy's expected accuracy estimate from separately measured realized accuracy, retain hats, and publish the component estimates and evaluation provenance.

e) Why the sensitivity axes matter together: The scorer defines correctness; the coupling defines the joint event; the pool changes both the union and the competing ceiling. Finite k instead changes the empirical marginals supplied to nonlinear estimators, not the population estimand. These sensitivity sets have different constructions and should not be ranked as if their ranges estimated a common population effect. Reporting

only one axis can leave the other choices hidden.

f) Reporting an interpretable comparison: Table IX lists disclosures that help separate the protocol, decision classes, target quantities, and empirical evidence. It is not a guarantee that a comparison is validated. In particular, a normalized router gap needs a named, independently evaluated policy; neither the hindsight union nor the clairvoyant ceiling can supply its accuracy. The historical archive, outcome-informed scorer development, and other limits in Section VIII remain constraints on the present evidence.

VIII. LIMITATIONS

First, the responses are historical and exact model/tokenizer revisions were not all captured; this is not a current leaderboard or a generation reproduction. Second, the fixed benchmarks and model pool were informed by prior analysis of these archived tensors, so their selection was not outcome-blind. They are case studies, not random samples, and the analysis has no confidence intervals, p -values, or cross-benchmark average. Third, product is the joint law of specified independent generation and of the independent-index empirical resampling experiment; neither certifies the historical cross-model generation mechanism. Same-index pairing is an engineering convention. Zero premium is attainable when only marginals are fixed and couplings may vary, not a competing value within an already specified independent joint protocol.

Fourth, the post-output-developed GSM8K and GPQA v2 scorers have only the limited post-hoc human check in Appendix D, not an outcome-blind development holdout. It leaves 4/30 GSM8K and 19/30 GPQA-Diamond cells without definite two-person consensus. Agreement on the determined subsets does not establish archive-wide scorer accuracy, validate benchmark references, or exclude shared human error. The

Exact sensitivity over every nonempty subpool

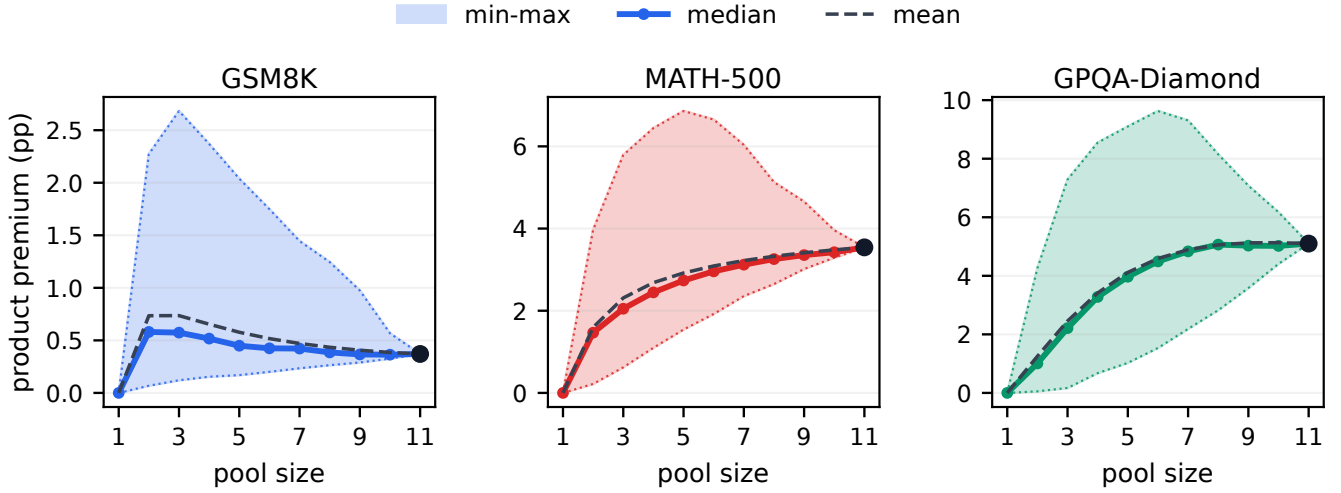


Fig. 8. Display-channel product premiums (pp) over every nonempty subpool, grouped by size. Shading is the exact minimum–maximum envelope of the enumerated set, not sampling uncertainty; black points mark the full pool.

TABLE IX

RECOMMENDED REPORTING CONTRACT FOR STOCHASTIC ROUTING BENCHMARKS. THE ROWS SEPARATE INPUTS, ESTIMANDS, SENSITIVITY AXES, AND THE EXTRA EVIDENCE NEEDED FOR ROUTER-DEPENDENT CLAIMS.

Contract item	Minimum disclosure
Protocol identity	Ordered queries and models; prompt; exact revisions; decoder; response count; scorer implementation and fingerprint.
Decision class	Whether selection occurs before or after seeing responses, and the model-call and verifier budget available to each comparator.
Target quantity	Absolute single-commit ceiling, named-coupling union, and their premium before reporting a normalized gap.
Joint assumption	The cross-model coupling used for a union, together with the marginally identified Fréchet interval.
Evaluation policy	Every material scorer channel and whether its policy was fixed before output inspection or developed after it.
Sensitivity	Declared model pools and deterministic finite-draw paths, including when and how each was fixed; their ranges must not be presented as confidence intervals.
Router-dependent claim	An independently evaluated policy q , with its training split, information set, and evaluation protocol.
Artifact lineage	Raw responses where releasable, identity manifests, machine-readable results, hashes, and a deterministic number-to-table path.

small, score-stratified sample and the decision to forgo follow-up after inspecting the initial-judgment summary further limit the check. Reporting all retained scorer channels reveals sensitivity but does not validate extraction semantics; the original release reproduces frozen score-tensor-to-result computation, not the entire raw-text-to-score path. Fifth, $k = 30$ is finite. The prefix analysis does not prove stationarity, exchangeability, or population convergence. Sixth, exact enumeration removes subpool-selection opacity only for this 11-model set; its extrema are outcome-descriptive and do not generalize to unseen models. The additional policy illustration holds out evaluation scores from the current fitting and tuning procedure but reuses this historically inspected archive; it does not establish a pristine holdout or new-data generalization. Finally, the single-commit ceiling excludes cascades, parallel ensembles, response-aware

selection, and verifier-assisted sampling, and the product union uses more model responses than a one-call policy.

IX. CONCLUSION

Per-model success probabilities do not, in general, determine a hindsight any-correct union under stochastic decoding. Proposition 1 identifies the sharp interval, and Proposition 2 separates a named product union from the ceiling for pre-response single-commit policies. These results distinguish an identification question from finite-sample estimation and from the performance of a learned router.

In three fixed historical tensors, the named product coupling yields premiums of 0.371, 3.542, and 5.100 percentage points on the display channels, with scorer ranges of 0.359–0.398, 3.542–3.754, and 5.100–6.632 points. For each benchmark, the

zero lower endpoint follows from the Fréchet identification result over couplings with those marginals. It does not negate a positive premium under a specified independent joint protocol. Exact subpool enumeration and frozen finite- k paths further show that neither full-pool maximality nor smooth convergence should be assumed. In these display-channel archives, GPQA-Diamond’s larger mean product premium than GSM8K’s reflects a larger positive-premium fraction, not a larger conditional mean among those queries.

The practical lesson is to report the conditional premium and its identification interval, then make the scorer, pool, and finite-draw checks explicit. For a reader with an independently evaluated router, the decomposition also supplies a way to state which part of its product-reference gap depends on that policy. The archived policy illustration instantiates that comparison without establishing a routing improvement or historical data independence. The later human check supports scorer agreement only on its determined subsets; unresolved cases, especially on GPQA-Diamond, and historical output exposure remain limitations.

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APPENDIX A

PROOFS AND COUNTEREXAMPLES

A. Proof of Proposition 1

Let $E_{im} = \{B_{im} = 1\}$. Since $E_{im} \subseteq \cup_{\ell} E_{i\ell}$ for every m ,

$$U_i^\mu = \Pr(\cup_m E_{im}) \geq \max_m \Pr(E_{im}) = R_i.$$

The union bound and the fact that a probability is at most one give $U_i^\mu \leq \min(1, \sum_m p_{im})$.

For attainability, let Z be uniform on the unit circle. Nested arcs $E_{im} = [0, p_{im})$ attain the lower endpoint R_i . For the upper endpoint, place arcs of lengths p_{im} consecutively around the

circle; their union has measure $\min\{1, \sum_m p_{im}\}$. In either construction, the induced law of $(\mathbf{1}\{Z \in E_{i1}\}, \dots, \mathbf{1}\{Z \in E_{iM}\})$ is a valid coupling on $\{0, 1\}^M$ with the required marginals.

Let μ_i^L and μ_i^H denote the lower- and upper-attaining couplings. For $t \in [0, 1]$, the mixture $\mu_{i,t} = (1-t)\mu_i^L + t\mu_i^H$ retains those marginals. Because $U_i^\mu = 1 - \mu_i(0, \dots, 0)$ is affine in μ_i ,

$$U_i^{\mu_{i,t}} = (1-t)R_i + t \min\left\{1, \sum_m p_{im}\right\}.$$

Thus every value in the interval is attainable. Since μ_i is specified separately for each query, endpoint-attaining couplings may be chosen querywise before equal-weight averaging; hence the averaged endpoints are attainable. Using the same mixture weight t for every query sweeps every value between those averaged endpoints, so the averaged identified set is the full interval. The product coupling is admissible and therefore $\Delta_i^\perp \geq 0$. $\square$

B. Proof of Proposition 2

The law of total probability gives

$$\begin{aligned} \Pr(B_{iM_i} = 1) &= \sum_{m: \pi_i(m) > 0} \Pr(M_i = m) \\ &\quad \cdot \Pr(B_{im} = 1 \mid M_i = m), \end{aligned}$$

Conditional independence of the pre-response choice and the fresh scoring response then gives $\Pr(B_{im} = 1 \mid M_i = m) = p_{im}$ on this support. Restoring the omitted zero-mass terms, the display equals $\sum_m \pi_i(m)p_{im} = q_i$. Because π_i lies on the probability simplex,

$$q_i = \sum_m \pi_i(m)p_{im} \leq \sum_m \pi_i(m)R_i = R_i.$$

Adding and subtracting $\bar{R}$ from $\bar{U}^\perp - \bar{q}$ proves Eq. (9). Non-negativity follows from the pointwise policy ceiling and Proposition 1. $\square$

C. Proof of Corollary 1

Let $d = \bar{\Delta}^\perp > 0$. Then $s(q) = d/(\bar{U}^\perp - \bar{q})$ and $\bar{U}^\perp = \bar{R} + d$. For $\bar{q} \leq \bar{R}$ the denominator is positive and strictly decreases as $\bar{q}$ increases, so $s(q)$ strictly increases. $\square$

D. Product-pool increments

For a pool A and a new model $\ell \notin A$,

$$U_{A \cup \{\ell\}}^\perp - U_A^\perp = p_\ell(1 - U_A^\perp) \geq 0.$$

Thus the product union is monotone with diminishing returns. Since $R_{A \cup \{\ell\}} - R_A = (p_\ell - R_A)_+$,

$$\Delta_{A \cup \{\ell\}}^\perp - \Delta_A^\perp = p_\ell(1 - U_A^\perp) - (p_\ell - R_A)_+.$$

The premium therefore need not be monotone. For A with marginals $(0.5, 0.5)$, $\Delta_A^\perp = 0.25$; adding a model with marginal one makes both union and ceiling one, and the premium becomes zero.

E. Proof of Proposition 3

At $k = 1$, every empirical marginal is zero or one. If all are zero, both product union and maximum are zero; otherwise both are one. Hence the premium is zero. At the full draw count, every permutation contains the same multiset of indices, so all empirical marginals and derived deterministic functionals coincide.

Nested prefixes do not imply monotonicity. For model-1 and model-2 draw sequences $b_{1,1:3} = (1, 0, 1)$ and $b_{2,1:3} = (0, 1, 1)$ in that order, the premiums at $k = 1, 2, 3$ are 0, $1/4$, and $2/9$, respectively. $\square$

F. Conditional population bias result

This result is not assumed by the fixed-tensor headline. Suppose nevertheless that each cell contains k iid Bernoulli responses with parameter p_{im} and that cell streams are independent across models for a fixed query. Then

$$\mathbb{E}[\hat{U}_i^\perp] = U_i^\perp, \quad 0 \leq \mathbb{E}[\hat{R}_i] - R_i \leq \sqrt{\frac{\log M}{2k}},$$

and

$$-\sqrt{\frac{\log M}{2k}} \leq \mathbb{E}[\hat{\Delta}_i^\perp] - \Delta_i^\perp \leq 0.$$

The first identity follows by factorizing $\mathbb{E}[\prod_m (1 - \hat{p}_{im})]$. Jensen's inequality gives the lower maximum-bias bound. Hoeffding's lemma yields $\mathbb{E} \exp\{\lambda(\hat{p}_{im} - p_{im})\} \leq \exp\{\lambda^2/(8k)\}$; log-sum-exp and optimization at $\lambda = \sqrt{8k \log M}$ give the upper bound for $M \geq 2$. When $M = 1$, $\hat{R}_i = \hat{p}_{i1}$ is unbiased and $U_i^\perp = R_i$, so both relevant biases are zero. Subtracting the nonnegative maximum bias from the unbiased union establishes the premium statement. At $M = 11, k = 30$ the bound is $\sqrt{\log(11)/60} = 0.1999$, far wider than the empirical premiums and not a useful data-specific correction. None of these population expectations is inferred merely from the archived same-index layout.

APPENDIX B

ADDITIONAL ARCHIVED-TENSOR DIAGNOSTICS

A. Same-index coupling diagnostic

For a draw prefix J , let a single index be sampled uniformly and reused across models. The resulting empirical coupling has marginals $\hat{p}_{im}(J)$, so its union lies inside the empirical Fréchet bracket. Its difference from product may have either sign: for draw sequences $b_{1,1:2} = (1, 0)$ and $b_{2,1:2} = (1, 0)$ the paired/product unions are $0.5/0.75$; for $b_{1,1:2} = (1, 0)$ and $b_{2,1:2} = (0, 1)$ they are $1/0.75$. A small difference therefore neither tests nor validates independence.

B. Per-model descriptive accuracies and a best-fixed anchor

Table XI reports per-model means for the display channels. For each benchmark and scorer separately, define

$$\begin{aligned} \hat{A} &= \max_m \frac{1}{N} \sum_{i=1}^N \hat{p}_{im}, \\ \hat{G} &= \hat{R} - \hat{A}, \\ \hat{R} &= \frac{1}{N} \sum_{i=1}^N \max_m \hat{p}_{im}. \end{aligned}$$

Here $\hat{A}$ is the largest model column mean, selected and measured on the same archive. In contrast, $\hat{R}$ takes the maximum within each query before averaging. Every model's column mean is bounded by the mean of the querywise maxima, so $\hat{G} \geq 0$. The difference quantifies the effect of these two aggregation orders in the fixed tensor.

This comparison was added in response to review, after the original results were known. Table X uses all retained scorer channels with the original full pool and full set of archived draws. On the display channels, the differences are 2.660 pp for GSM8K, 5.087 pp for MATH-500, and 32.273 pp for GPQA-Diamond; the full-pool product premiums on those same channels are 0.371, 3.542, and 5.100 pp, respectively (Table III). The anchor is selected and measured in-sample, so it is a selection-optimistic descriptive reference rather than held-out policy performance. It is not an independently evaluated $\bar{q}$ in Eq. (9); the difference $\hat{G}$ does not establish an attainable routing improvement. The product premiums in Tables III and V compare the product union with the querywise ceiling instead of comparing aggregation orders.

C. Complete named-pool sensitivity

Table XII expands the leave-one-declared-group-out portion of the fixed-pool audit. Each row changes composition and pool size together: the Qwen2.5 removal drops three models, whereas each other row drops one. The rows are descriptive joint contrasts and do not isolate composition, pool size, or group-level causal effects.

D. Complete finite-draw path summaries

Table XIII reports every frozen k used in the analysis. The shared $k = 30$ endpoints are exact full-archive identities, while the intermediate min-max brackets summarize order sensitivity across the eight precomputed permutations.

E. Submitted evidence and revision additions

The original submitted bundle contains the frozen tensors, numerical contract, canonical Run-A outputs, analysis receipt, and per-member hash manifest. It omits analysis source, the release-control files listed in Section V-D, raw responses, and the runtime. The original frozen input manifest, numerical contract, scientific results file, and submitted ZIP have SHA-256 prefixes a7c08b3a, 1e334469, b95afa09, and 76e3fdf2, respectively. These identify the original submission, not the additions below.

The post-review computations in Section VI-A and Appendix B-B are recorded separately in `descriptive_results.json`, with SHA-256 prefix 01f6fef1. The associated querywise records and analysis receipt retain the frozen-input bindings. These new outputs are not contained in the original submitted ZIP and must accompany a revised evidence package.

Appendix C uses a separate revision protocol and result record, rather than relabeling the original no-policy-output release. The protocol, evaluation result, and independent arithmetic-verification record have SHA-256 prefixes

TABLE X

IN-SAMPLE AGGREGATION-ORDER COMPARISON ON THE FROZEN SCORES, USING 11 MODELS AND 30 ARCHIVED DRAWS PER CELL. A DAGGER MARKS THE EXISTING DISPLAY CHANNEL. THE FIRST TWO NUMERICAL COLUMNS ARE EMPIRICAL SCORE LEVELS (%); THEIR DIFFERENCE $\hat{G}$ IS IN PERCENTAGE POINTS (PP).

Scorer	Querywise ceiling $\hat{R}$ (%)	Best column mean $\hat{A}$ (%)	Difference $\hat{G}$ (pp)
GSM8K			
gsm8k_v2 [†]	98.980	96.320	2.660
gsm8k_v1	98.887	95.880	3.007
gsm8k_legacy	98.893	96.053	2.840
MATH-500			
math500_mathverify [†]	87.713	82.627	5.087
math500_legacy	83.687	77.267	6.420
GPQA-Diamond			
gpqa_v2 [†]	82.694	50.421	32.273
gpqa_v1	84.680	45.673	39.007
gpqa_legacy	87.778	50.572	37.205

Differences are computed from unrounded exact values and then rounded; subtracting the printed levels can differ in the last decimal place.

TABLE XI

PER-MODEL EMPIRICAL ACCURACY (%) ON DISPLAY CHANNELS: MEAN BINARY SCORE OVER EACH BENCHMARK'S FIXED QUERIES AND 30 ARCHIVED DRAWS PER CELL.

Model	GSM8K	MATH-500	GPQA-Diamond
Mistral-7B-Instruct-v0.3	53.460	14.927	13.199
DeepSeek-R1-Distill-Qwen-7B	85.907	53.687	15.354
Qwen2.5-7B-Instruct-AWQ	91.447	74.393	28.300
Qwen2.5-14B-Instruct-AWQ	93.640	77.740	35.859
Qwen2.5-32B-Instruct-AWQ	96.320	82.627	38.199
Phi-4	94.647	80.560	50.421
OLMo-2-1124-7B-Instruct	79.973	36.240	26.734
Yi-1.5-9B-Chat	86.780	51.427	28.569
Granite-3.3-8B-Instruct	89.113	69.273	28.401
Gemma-2-9B-IT	87.387	48.980	22.071
Llama-3.1-8B-Instruct	85.613	49.440	23.232

710bf924, 2e7c3b42, and 44561e0e, respectively. The result includes exact fractions for the matched empirical decomposition. The verification checks its arithmetic and record bindings; it is not an independent refitting of the policy or evidence of historical data independence.

APPENDIX C

A HISTORICAL-ARCHIVE ROUTER ILLUSTRATION

This post-review illustration supplies a policy-specific comparator for Eq. (9), using the existing MATH-500 archive rather than new model responses. It uses all 500 queries, the same 11-model pool, and 30 archived draws per query-model cell under the Math-Verify display channel. The held-out designation applies to the fitting and selection procedure below, not to the author's historical exposure to the archive.

a) Policy and split: A fixed seeded split allocated 300 queries to TRAIN, 100 to TUNE, and 100 to EVAL. The split kept connected components of token-Jaccard similarity at threshold 0.70 within one role; this heuristic does not exclude every semantic near-duplicate. No query was excluded based on scores or premium. The router uses TF-IDF query features and cosine nearest neighbors. Vocabulary, inverse document frequencies, and model success rates were fitted

only on TRAIN. TUNE selected the neighbor count from $\{5, 10, 25, 50\}$, yielding $K = 25$, without refitting on TUNE. Ties use fixed orders, with the smaller K preferred for a tuning tie. The fixed-model comparator was also selected only on TRAIN: Qwen2.5-32B-Instruct-AWQ. The fitting process could read TRAIN and TUNE scores but not EVAL scores; the prediction process received the fitted policy and EVAL query text, not EVAL scores. Its deterministic model choices were sealed before the separate evaluation stage read those scores. The run used CPU computation, with no new response generation.

b) Matched empirical quantities: For each EVAL query, let $a(i)$ be the sealed model choice and $\hat{p}_{im} = 30^{-1} \sum_{j=1}^{30} B_{imj}$. All quantities in Table XIV use these same marginals, queries, scorer, pool, and equal query weights. In particular,

$$\hat{q} = \frac{1}{100} \sum_{i \in \text{EVAL}} \hat{p}_{i, a(i)}.$$

The ceiling $\hat{R}$ and product union $\hat{U}^\perp$ use the within-query maximum and product of Section IV before averaging over EVAL. Here $\hat{q}$ is the mean archived success rate of the selected models, equivalently the expected accuracy under uniform resampling of an archived draw for each selection. It is not

TABLE XII

DISPLAY-CHANNEL PRODUCT PREMIUMS (PP) AFTER EACH DECLARED GROUP REMOVAL FROM THE FIXED 11-MODEL POOL. THESE CONTRASTS CHANGE COMPOSITION AND SIZE TOGETHER; THEY DO NOT ISOLATE FAMILY EFFECTS.

Removed group	GSM8K	MATH-500	GPQA-Diamond
Mistral	0.326	3.460	4.482
DeepSeek-Distill-Qwen	0.327	3.500	4.885
Qwen2.5	0.404	3.633	7.729
Phi	0.567	3.648	5.452
OLMo	0.366	3.423	5.022
Yi	0.364	3.403	5.158
Granite	0.352	3.311	4.408
Gemma	0.459	3.516	4.919
Llama	0.360	3.364	4.420

TABLE XIII

DISPLAY-CHANNEL FINITE- k SUMMARIES (PP): MEAN [MINIMUM, MAXIMUM] ACROSS EIGHT FIXED PERMUTATIONS. INTERMEDIATE MEANS DIAGNOSE PREFIX SENSITIVITY; WITHIN- k RANGES DIAGNOSE ORDER SENSITIVITY, NOT UNCERTAINTY COVERAGE. THE $k = 1$ ZERO AND SHARED $k = 30$ ENDPOINT ARE STRUCTURAL, NOT CONVERGENCE EVIDENCE.

k	GSM8K	MATH-500	GPQA-Diamond
1	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
2	0.066 [0.000, 0.200]	1.124 [0.925, 1.325]	1.877 [1.326, 3.212]
5	0.236 [0.166, 0.349]	2.372 [2.172, 2.502]	3.559 [2.990, 4.201]
10	0.317 [0.238, 0.421]	3.024 [2.873, 3.202]	4.398 [3.796, 4.941]
15	0.337 [0.269, 0.380]	3.251 [3.052, 3.396]	4.662 [4.151, 5.078]
20	0.349 [0.312, 0.393]	3.398 [3.208, 3.518]	4.757 [4.505, 5.127]
25	0.358 [0.339, 0.380]	3.478 [3.403, 3.533]	4.909 [4.610, 5.185]
30	0.371 [0.371, 0.371]	3.542 [3.542, 3.542]	5.100 [5.100, 5.100]

TABLE XIV

ONE MATCHED EVAL SPLIT OF THE HISTORICAL MATH-500 ARCHIVE. THE FIXED COMPARATOR IS SELECTED ON TRAIN, UNLIKE THE IN-SAMPLE ANCHOR IN APPENDIX B-B. LEVELS ARE PERCENTAGES, DIFFERENCES ARE PERCENTAGE POINTS (PP), AND SHARES ARE UNITLESS. VALUES ARE ROUNDED ONLY FOR DISPLAY.

Quantity	TF-IDF/KNN router	TRAIN-selected fixed
$\hat{q}$ (%)	79.6667	80.0333
$\hat{R}$ (%)	85.0000	85.0000
$\hat{U}^\perp$ (%)	88.2826	88.2826
$\hat{\Delta}^\perp$ (pp)	3.2826	3.2826
$\hat{R} - \hat{q}$ (pp)	5.3333	4.9667
$\hat{U}^\perp - \hat{q}$ (pp)	8.6160	8.2493
$\hat{s}$	0.3809943	0.3979288

accuracy measured on fresh routed calls. The fixed comparator uses its TRAIN-selected model for every EVAL query. The premium is $\hat{\Delta}^\perp = \hat{U}^\perp - \hat{R}$, and the share divides this premium by $\hat{U}^\perp - \hat{q}$, which is positive for both comparators. The product union is the conditional union probability for independent per-model archived-draw resampling, not the observed same-index union or a certification of the historical generation coupling.

The router did not outperform the TRAIN-selected fixed model on this split. The example nevertheless makes the decomposition concrete: the same premium has a different share when the comparison policy changes, as in Corollary 1. Neither share is a fraction of guaranteed attainable routing improvement, and the ceiling gap is not shown to be learnable.

This is a single descriptive illustration, not a new routing method or a claim about performance on other queries.

c) Historical exposure and evidence scope: The benchmark, pool, and outcomes had already been inspected during the paper’s development, and all 500 gold answers had been used in scorer parsing and self-equivalence checks. Data preparation and the host administrator could access the complete selected score channel. Restricting the fitting and prediction stages therefore does not turn this archive into a previously unseen holdout or establish historically independent research choices. The matched inequalities $\hat{q} \leq \hat{R} \leq \hat{U}^\perp$ and $\hat{s} \in [0, 1]$ are algebraic checks, not tests for leakage. No population confidence interval or deployment-performance inference is claimed. The separate post-review result record has SHA-256 prefix 2e7c3b42; it does not alter the original release’s absence of named policy outputs.

APPENDIX D

POST-REVIEW REFERENCE-BASED HUMAN CHECK

a) Scope and sampling: This check compares human judgments of archived final answers with the existing GSM8K and GPQA-Diamond frozen-v2 scores. It does not re-score the archive or assess MATH-500. Before sampling, the proposed workload was reduced from 100 to 30 response cells per benchmark to keep unpaid participation manageable. The resulting 60 distinct cells contain 15 from each frozen-v2 score side per benchmark. Selection excluded documented scorer-development queries, used a fixed salted model rotation, and prohibited repeated queries within each benchmark. This

TABLE XV

HUMAN CHECK OF THE EXISTING FROZEN-V2 SCORES. MATCH AND MISMATCH REQUIRE TWO MATCHING, DEFINITE INITIAL HUMAN JUDGMENTS; UNDETERMINED INCLUDES EVERY OTHER PAIR. CONDITIONAL MATCH DIVIDES MATCHES BY THE NUMBER WITH A DEFINITE CONSENSUS, NOT BY N . ALL SAMPLED CELLS ARE RETAINED, AND NO FOLLOW-UP JUDGMENT WAS COLLECTED. THESE ARE COUNTS FOR A FIXED STRATIFIED SAMPLE, NOT ESTIMATES OF ARCHIVE-WIDE SCORER ACCURACY.

Benchmark	Frozen-v2 side	N	Match	Mismatch	Undetermined	Conditional match
GSM8K	All	30	26	0	4	26/26
	0	15	12	0	3	12/12
	1	15	14	0	1	14/14
GPQA-Diamond	All	30	11	0	19	11/11
	0	15	4	0	11	4/4
	1	15	7	0	8	7/7

is a score-stratified descriptive sample, not a uniform sample of the archive. Excluding known development cases cannot establish that all prior exposure has been identified.

b) Task and blinding: Three unpaid contributors each received 40 items: 20 from each benchmark. Every cell received two initial judgments, and each contributor pair shared 20 cells. The packets showed the question, its reference answer, and the raw response, withholding automated decisions, model-identifying metadata, and other contributors' judgments. Contributors were instructed to work separately, without AI or automated scoring tools, and to record a supported final-answer span and a reason. The common rules allowed CORRECT, INCORRECT, and UNRESOLVED; reference ambiguity, uncertain record completeness, and expertise limits were reasons to remain unresolved. The task was final-answer equivalence to the supplied reference, not a full proof audit or independent verification of benchmark gold. Procedural blinding alone does not establish evaluator independence or expertise across every GPQA subject; unchanged raw responses can also contain model clues.

c) Closure and a change to the planned procedure:

The original procedure allowed up to ten follow-up judgments per contributor. After receiving the 120 initial judgments and inspecting their agreement summary, but before comparing them with automated scores, the author decided not to request further annotation, changing the planned procedure. Matching CORRECT/CORRECT or INCORRECT/INCORRECT pairs define an analysis consensus; all other pairs remain undetermined, with both original labels retained. The closure rule and resulting consensus records were locked before the scorer comparison. No follow-up, third-person adjudication, majority vote, or author/AI replacement judgment was used. All 60 cases remain reported; conditional agreement is restricted to the definite-consensus subset.

d) Results: Table XV reports the comparison separately by benchmark and original score stratum. On GSM8K, the initial pairs comprise 26 matching definite judgments, three with exactly one UNRESOLVED, and one with both judgments UNRESOLVED. On GPQA-Diamond, the corresponding counts are 11, 15, and three, plus one CORRECT/INCORRECT disagreement. Opposite judgments and two unresolved judgments both remain undetermined, not definite matches or mismatches with the scorer.

e) Interpretation and limits: No scorer disagreement was observed among definite-consensus cases, but GPQA-Diamond's low coverage prevents a broad validation claim. Returned reasons include subject-matter uncertainty, doubts about response completeness, and question/reference ambiguity; the definite disagreement also reflects different emphases on the final option and intermediate reasoning. This describes existing reasons, not new adjudications or grounds for dropping cases.

The determined subset cannot represent the unresolved cases or exclude shared human mistakes. We report these counts without a population estimate or confidence interval. This check provides partial evidence for these two frozen-v2 channels, not validation of all models, channels, or historical scorer-development choices.